



The Quadratic Formula

You can solve any quadratic equation in standard form using the quadratic formula.

Use completing the square on the standard form of a quadratic equation, $ax^2 + bx + c = 0$, to derive the **quadratic formula**.

$$ax^2 + bx + c = 0$$

$$ax^2 + bx = -c$$

Subtract c from each side.

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Divide each side by a .

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2}$$

Add $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2}$ to each side.

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{4ac}{4a^2} + \frac{b^2}{4a^2}$$

Multiply $\frac{c}{a}$ by $\frac{4a}{4a}$ to get a common denominator.

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Simplify.

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Factor the perfect square trinomial.

$$\left(x + \frac{b}{2a}\right) = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

Take the square root of each side.

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Simplify.

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Subtract $\frac{b}{2a}$ from each side.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Simplify.

Using the Quadratic Formula

Quadratic Formula

For a quadratic equation written in standard form, $ax^2 + bx + c = 0$, where $a \neq 0$, the solutions of the equation can be found using the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To use the quadratic formula, write the quadratic equation you are trying to solve in standard form, and then identify the values of a , b , and c .

EXAMPLE 1

Solve the equation. Estimate irrational solutions to the nearest tenth.

A $x^2 + 9x - 22 = 0$

SOLUTION

Since the equation is in standard form, identify a , b , and c : $a = 1$, $b = 9$, and $c = -22$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-9 \pm \sqrt{9^2 - 4 \cdot 1 \cdot (-22)}}{2 \cdot 1}$$

Substitute values for a , b , and c .

$$x = \frac{-9 \pm \sqrt{81 - (-88)}}{2}$$

Simplify the radicand.

$$x = \frac{-9 \pm \sqrt{169}}{2}$$

Simplify.

$$x = \frac{-9 \pm 13}{2}$$

Evaluate the square root.

$$x = \frac{-9 + 13}{2} \text{ or } x = \frac{-9 - 13}{2}$$

Write $x = \frac{-9 \pm 13}{2}$ as two equations.

$$x = \frac{4}{2} \qquad x = \frac{-22}{2}$$

Simplify each numerator.

$$x = 2 \qquad x = -11$$

Divide.

The solution set is $\{2, -11\}$.

B $x^2 + 3x = 5$

SOLUTION

Write the equation in standard form by subtracting 5 from each side.

$$x^2 + 3x = 5$$

$$x^2 + 3x - 5 = 0 \quad \text{Subtract 5 from each side.}$$

Identify a , b , and c : $a = 1$, $b = 3$, and $c = -5$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-5)}}{2 \cdot 1}$$

$$x = \frac{-3 \pm \sqrt{9 - (-20)}}{2}$$

$$x = \frac{-3 \pm \sqrt{29}}{2}$$

$$x = \frac{-3 + \sqrt{29}}{2} \approx 1.2 \text{ or } x = \frac{-3 - \sqrt{29}}{2} \approx -4.2$$

Substitute values for a , b , and c .

Simplify the radicand.

Simplify.

Write $x = \frac{-3 \pm \sqrt{29}}{2}$ as two equations.

The solution set is $\{1.2, -4.2\}$. ■

Using the Discriminant

The **discriminant** is the radicand $b^2 - 4ac$ in the quadratic formula. If you know the value of the discriminant, you can determine how many solutions the equation has and whether the solutions are rational, irrational, or nonreal.

Property

The equation, $ax^2 + bx + c = 0$, where $a \neq 0$, has

- Two rational solutions if $b^2 - 4ac > 0$ and $b^2 - 4ac$ is a perfect square.
- Two irrational solutions if $b^2 - 4ac > 0$ and $b^2 - 4ac$ is not a perfect square.
- One rational solution if $b^2 - 4ac = 0$.
- No real solutions if $b^2 - 4ac < 0$.

EXAMPLE 2

Use the discriminant to determine the number and type of solutions the equation has.

► **Remember** A number is a perfect square if it can be written as the square of a rational number.

A $2x^2 + 13x + 15 = 0$

SOLUTION

Identify a , b , and c : $a = 2$, $b = 13$, and $c = 15$.

$$b^2 - 4ac = 13^2 - 4 \cdot 2 \cdot 15 = 169 - 120 = 49$$

The discriminant, 49, is positive and is a perfect square, so the equation has two rational solutions.

B $x^2 - \frac{1}{2}x + \frac{1}{16} = 0$

SOLUTION

Identify a , b , and c : $a = 1$, $b = -\frac{1}{2}$, and $c = \frac{1}{16}$.

$$b^2 - 4ac = \left(-\frac{1}{2}\right)^2 - 4 \cdot 1 \cdot \frac{1}{16} = \frac{1}{4} - \frac{1}{4} = 0$$

The discriminant is zero, so the equation has one rational solution.

C $x^2 + 5 = 0$

SOLUTION

Identify a , b , and c : $a = 1$, $b = 0$, and $c = 5$.

$$b^2 - 4ac = 0^2 - 4 \cdot 1 \cdot 5 = 0 - 20 = -20$$

The discriminant, -20 , is negative, so the equation has no real solutions. ■

Application: Determining Whether a Polynomial Is Factorable

The expression $ax^2 + bx + c$ is factorable if the solutions to the equation $ax^2 + bx + c = 0$ are rational. The expression $ax^2 + bx + c$ is factorable if the discriminant is either a perfect square or zero.

EXAMPLE 3

Determine whether the polynomial is factorable.

A $30x^2 - 31x - 44$

SOLUTION

Find the discriminant.

$$\begin{aligned}a &= 30, b = -31, \text{ and } c = -44 \\b^2 - 4ac &= (-31)^2 - 4 \cdot 30 \cdot (-44) \\&= 961 + 5280 \\&= 6241\end{aligned}$$

Because $6241 = 69^2$, it is a perfect square, so $30x^2 - 31x - 44$ is factorable.

B $27x^2 + 14x - 15$

SOLUTION

Find the discriminant.

$$\begin{aligned}a &= 27, b = 14, \text{ and } c = -15 \\b^2 - 4ac &= 14^2 - 4 \cdot 27 \cdot (-15) \\&= 196 + 1620 \\&= 1816\end{aligned}$$

Because 1816 is not a perfect square, $27x^2 + 14x - 15$ is not factorable. ■